Constraining the Standard Codon-to-Amino-Acid Assignment by WToken Fine Classes and Position-Weighted Cost

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Abstract

We present a constrained derivation of the standard genetic code's codon-to-amino-acid assignment. The 8-tick DFT basis of Recognition Science generates exactly 20 semantic atoms (WTokens), which are placed in bijection with the 20 canonical amino acids. The WToken mode/phase/offset structure partitions the amino acids into 8 fine candidate classes, giving an upper bound of 36,864 fine-class-respecting assignments. However, these fine classes do not refine the standard code's degeneracy pattern. When both constraints are imposed simultaneously, the admissible search space collapses to an exact 48 assignments, and the standard code is the unique optimum on this 48-state space under the position-weighted mutation cost of a companion paper. Both the 48-state count and the strict optimality theorem are machine-checked in Lean 4. This shows that the combination of WToken structure, observed degeneracy pattern, and position-weighted mutation cost is sufficient to single out the standard code from a tiny explicitly characterized residual space.

1 Introduction

A global-optimality result by itself does not explain why the standard genetic code assigns *these* 20 amino acids to *these* codon families. A derivation must do more than say that the code is good under a cost function; it must also constrain the admissible assignment space.

Recognition Science provides exactly such a constraint. The 8-tick DFT structure generates 20 neutral semantic atoms (WTokens), and the canonical WToken $\leftrightarrow$ amino-acid bijection partitions the amino acids into eight fine classes. At first glance this seems to reduce the search space dramatically: if assignments must respect the fine classes, then 20! collapses to an upper bound of $2!^6 \times 4!^2 = 36{,}864$.

The crucial subtlety is that the fine classes do *not* refine the standard degeneracy pattern. For example, Val and Leu lie in the same fine class but have 4-fold and 6-fold degeneracy respectively. Likewise Ser and Thr, Lys and Arg, and the four-member "special imaginary" class cross degeneracy boundaries. So the naive 36,864-state fine-class space is not the right object for an exact codon-assignment derivation.

This paper isolates the correct object: the intersection of

1. the WToken fine-class constraint, and
2. the observed standard-code degeneracy pattern.

That exact intersection has only 48 assignments, and we prove in Lean 4 that the standard code is the unique optimum on this 48-state space under the position-weighted mutation cost from the companion global-optimality paper [3].

2 The WToken Constraint

2.1 The 20 WTokens

The 8-tick DFT basis gives four mode families: modes 1+7, 2+6, 3+5, and the Nyquist mode 4. Together with four phase levels and the real/imaginary offset available only in mode 4, this yields exactly 20 WTokens:

$$3 \times 4 + 1 \times 4 \times 2 = 12 + 8 = 20.$$

In Lean, this is the theorem `wtoken_count` in `WTokenIso.lean`. The canonical amino acid type also has cardinality 20 (`amino_acid_count`), and the explicit map `wtoken_to_amino` is proved bijective (`wtoken_amino_bijection`).

2.2 Fine Candidate Classes

The canonical bijection induces eight fine classes:

Class	Members	WToken origin
Aliphatic small	Gly, Ala	mode 1+7, ϕ 0–1
Aliphatic branched	Val, Leu	mode 1+7, ϕ 2–3
Polar hydroxyl	Ser, Thr	mode 2+6, ϕ 0–1
Polar amide	Asn, Gln	mode 2+6, ϕ 2–3
Charged negative	Asp, Glu	mode 3+5, ϕ 0–1
Charged positive	Lys, Arg	mode 3+5, ϕ 2–3
Aromatic (real)	His, Phe, Tyr, Trp	mode 4, $\tau = 0$
Special (imaginary)	Pro, Cys, Met, Ile	mode 4, $\tau = 2$

This partition is formalized by `FineCandidateClass`, `aminoFineClass`, and `wtokenFineClass` in `WTokenAssignmentBridge.lean`; the theorem `canonical_bridge_respects_fineClass` proves that the canonical WToken $\leftrightarrow$ amino map respects these classes.

The resulting fine-class-respecting search space has size at most

$$2!^6 \times 4!^2 = 36{,}864,$$

formalized by `fineClassSearchBound_eq_factorials` in `NearUniqueness.lean`.

2.3 Why the Naive Fine-Class Space Is Not Enough

The eight fine classes are chemically plausible, but they do not line up with the standard code's degeneracy classes. Lean explicitly proves the basic counterexample:

$$\texttt{aminoFineClass Val} = \texttt{aminoFineClass Leu} \quad \text{but} \quad \texttt{degeneracyClass Val} \neq \texttt{degeneracyClass Leu}.$$

The theorem name is `sameFineClass_not_sameDegeneracy` in `WTokenFineClassOptimality.lean`.

This matters because any exact derivation of the standard code must respect the observed degeneracy pattern. A fine-class-respecting assignment can swap Val and Leu, but such a swap changes a 4-fold family into a 6-fold family and is therefore not an admissible reassignment of the actual code.

3 The Exact 48-State Intersection

3.1 Intersecting Fine Classes with Degeneracy

Once we impose both constraints simultaneously, only four nontrivial blocks remain:

- $\{\text{Gly}, \text{Ala}\}$ inside the 4-fold degeneracy class,
- $\{\text{Asn}, \text{Gln}\}$ inside the 2-fold class,
- $\{\text{Asp}, \text{Glu}\}$ inside the 2-fold class,
- $\{\text{His}, \text{Phe}, \text{Tyr}\}$ inside the 2-fold class.

All other amino acids are fixed because their fine classes cross degeneracy boundaries. The exact intersection count is therefore

$$2! \times 2! \times 2! \times 3! = 48.$$

This is formalized in Lean by the finite type `FineDegChoice` and the theorems

- `fineDegChoice_count` (`Fintype.card FineDegChoice = 48`),
- `fineDegChoice_count_factorials` (`2! × 2! × 2! × 3! = 48`),
- `choiceAssignment_respectsFineClass`,
- `choiceAssignment_respectsDegeneracy`.

3.2 Position-Weighted Cost on the 48-State Space

We now evaluate the exact 48-state space under the same position-weighted mutation cost used in the companion global-optimality paper [3]. That cost is defined in `PositionWeightedCost.lean` and uses the integer weight vector

$$(104, 55, 202, 4131, 4, 25, 1609, 3870),$$

corresponding to:

- hydrophobicity^2 at positions 1, 2, 3,
- charge^2 at position 3,
- side-chain volume^2 at positions 1, 2,
- biosynthetic pathway distance at positions 1, 2.

The exact theorem proved in Lean is:

Theorem (`standard_strictly_best_on_fineDegChoices`) For every non-identity `FineDegChoice` c,

$$\texttt{posStandardCost globalOptimalWeights} < \texttt{posTotalCos}$$

In words: among the 48 assignments that respect both the WToken fine classes and the standard degeneracy pattern, the standard code is the unique optimum.

3.3 Interpretation

This yields a clean two-stage derivation:

1. The 8-tick DFT structure forces 20 WTokens and an 8-class fine partition of the amino acids.
2. Intersecting that partition with the observed degeneracy pattern leaves exactly 48 admissible assignments.
3. The position-weighted mutation cost uniquely selects the standard code from those 48.

The derivation is therefore conditional in an important sense: it explains the standard code once the WToken structure and the observed degeneracy pattern are both imposed. It does not derive the degeneracy pattern itself.

4 Formal Verification

The structural pieces of this paper are formalized in Lean 4 (version 4.27.0) with Mathlib:

- `WTokenIso.lean`: the 20 WTokens, the explicit inverse, and the theorem `wtoken_amino_bijection`.
- `WTokenAssignmentBridge.lean`: the fine classes and the theorem `canonical_bridge_respects_fineClass`.
- `NearUniqueness.lean`: the upper bound `fineClassSearchBound = 36,864`.
- `WTokenFineClassOptimality.lean`: the exact 48-state intersection, the non-refinement counterexample `sameFineClass_not_sameDegeneracy`, and the theorem `standard_strictly_best_on_fineDegChoices`.
- `PositionWeightedCost.lean`: the position-weighted cost and the classwise certificates inherited from Paper 2.

The new module contains zero axioms and zero `sorry` assertions. Unlike the earlier overbroad version of this paper, the present draft does not rely on an invalid transfer from the 363,004 classwise search to the naive 36,864 fine-class space. The exact 48-state theorem is internal to Lean.

5 Discussion

5.1 Relation to Coevolution and Stereochemical Hypotheses

The derivation shares a structural feature with the coevolution hypothesis [5]: the WToken fine classes group amino acids by chemical and biosynthetic family, and the position-weighted cost gives large weight to biosynthetic pathway distance. The derivation goes beyond coevolution in two respects:

1. the classes are derived from the WToken DFT structure rather than being postulated from biochemistry, and
2. the exact 48-state residual space is fully enumerated and proved to have a unique optimum.

It is less aligned with the stereochemical hypothesis [4], since it does not posit direct codon–amino-acid affinity. The mechanism is structural rather than chemical: the WToken organization constrains the assignment space, and the mutation cost selects within that constrained space.

5.2 What Is and Is Not Derived

The derivation explains:

- why there are exactly 20 amino acids (= 20 WTokens),
- why those amino acids fall into the observed fine chemical families,
- why the standard code is the unique optimum once both WToken fine classes and the observed degeneracy pattern are imposed.

It does not explain:

- why the degeneracy pattern itself is what it is,
- why stop codons occupy their current positions,
- the physical mechanism connecting WToken mode structure to amino acid chemistry and tRNA recognition.

6 Conclusion

The exact codon-to-amino-acid assignment problem is not solved by WToken structure alone, but it is sharply constrained by it. The 8-tick DFT basis forces 20 WTokens and an 8-class fine partition of the amino acids; when that partition is intersected with the observed degeneracy pattern, only 48 assignments remain. The position-weighted mutation

cost then uniquely selects the standard code from this exact 48-state space.

This is a stronger and more precise result than the earlier overbroad claim about a naive 36,864-state fine-class derivation. The corrected theorem is fully formalized in Lean 4 and gives an exact constrained-derivation statement for the standard code.

Data Availability

All Lean source files are in the `reality` repository under `IndisputableMonolith/Genetics/` and `IndisputableMonolith/Water/WTokenIso.lean`.

References

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